Agent-Based Modeling of Renal Cell Carcinoma with Glucose Sensing

A Repast4Py Implementation

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Outline

1. Introduction & Motivation
2. Theoretical Core
3. Model Architecture
4. Key Systems
5. Implementation
6. Parameters & Optimisation
7. Results
8. Empirical Study
9. Conclusion

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Renal Cell Carcinoma — The Clinical Challenge

- 3rd most common urological cancer
- ∼400 000 new cases per year worldwide
- Metastatic 5-year survival $< 15\%$
- Clear-cell RCC (ccRCC) accounts for ∼75% of cases
- Strong sex differences:
  - Incidence ratio $\approx$ 2:1 (male : female)
  - Females show better immunotherapy response

Key Facts

Metric	Value
Incidence	400k/yr
M:F ratio	2:1
5-yr survival	<15%
Main subtype	ccRCC

Why Agent-Based Modeling?

Strengths of ABMs

- Natural representation of **heterogeneous** cell populations
- Capture **spatial dynamics** and local interactions
- **Emergent behaviour** arises from simple rules
- **In silico** experimentation — fast, ethical, cheap
- Stochasticity matches biological variability

Limitations of ODE/PDE Models

- Assume well-mixed populations
- Difficult to capture cell-level heterogeneity
- No spatial resolution
- Hard to model discrete events (mutation, death)

Project Objectives

1. **Port Mesa** → **Repast4Py**: migrate the existing RCC ABM to a scalable MPI-based framework
2. **Add glucose microenvironment**: 3D reaction–diffusion field modelling the Warburg effect
3. **Validate against clinical data**: fit model to ARON-1 clinical survival outcomes
4. **Bayesian optimisation**: tune 114 parameters with Optuna (TPE sampler, 250 trials)
5. **Sex-specific analysis**: compare male vs. female immune dynamics under treatment

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ABM Formalism & Repast4Py

ABM Components

Agents	autonomous entities with state & rules
Environment	3D grid with glucose overlay
Rules	step functions executed each tick
Emergence	macro-level tumour dynamics from micro-level interactions

Repast4Py Primitives

- `repast4py.core.Agent` — base agent
- `SharedContext` — distributed agent container
- `SharedGrid` — MPI-partitioned grid
- `BorderType.Sticky` — clamped boundaries
- `OccupancyType.Multiple` — co-location
- `mpi4py.MPI` — inter-rank communication

Immune System Overview

Innate Immunity

- **M1 Macrophages** — phagocytosis, pro-inflammatory
- **M2 Macrophages** — tissue repair, pro-tumour
- **NK Cells** — MHC-independent killing
- **Dendritic Cells (DC)** — antigen presentation
- **Plasmacytoid DC** — type I IFN production
- **Mast Cells** — angiogenesis modulation
- **Neutrophils** — early response

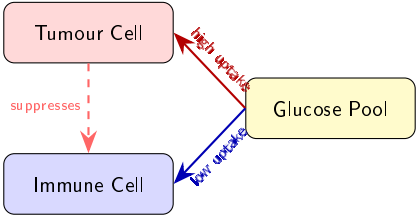
Adaptive Immunity

- **$CD8^+$ Cytotoxic T Cells** — direct tumour killing
- **$CD8^+$ Naïve T Cells** — precursors
- **$CD4^+$ Naïve T Cells** — helper precursors
- **Th1 Cells** — promote CTL & M1
- **Th2 Cells** — promote humoral response
- **Treg Cells** — immune suppression

Also modelled: **Adipocytes**, **Blood vessels**, **Sex hormones**, **Cytokines**

The Warburg Effect

- Tumour cells preferentially use **aerobic glycolysis** even in normoxia
- Glucose consumption rate ∼10–200× higher than normal cells
- Creates **metabolic competition** with immune cells
- Glucose-depleted microenvironment impairs:
  - T cell activation and proliferation
  - NK cell cytotoxicity
  - DC antigen presentation
- Glucose availability becomes a **limiting factor** for immune function



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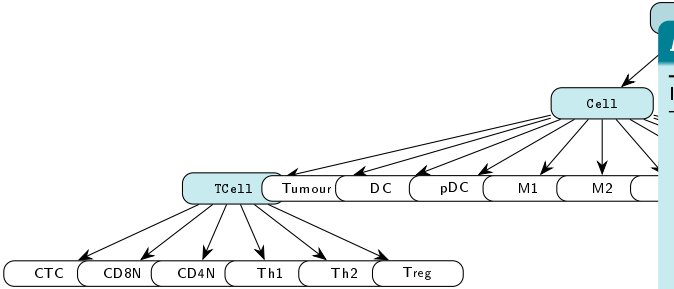
Agent Hierarchy — 18 Types



AgentType IntEnum

ID	Type
0	TumorCell
1	CD8 Cytotoxic
2	CD8 Naïve
3	CD4 Naïve
4	Th1
5	Th2
6	Treg
7	Dendritic Cell
8	Plasmacytoid DC
9	Macrophage M1
10	Macrophage M2
11	Natural Killer
12	Mast Cell
13	Neutrophil
14	Adipocyte
15	Blood
16	Sex Hormone
17	Cytokine

Interaction Matrix

Agent	Key Interactions
$CD8^+$ CTL	Kills tumour (TCR match + no PD-1 inhibition)
M1 Macrophage	Phagocytoses tumour, boosts T kill rate & Th1 proliferation
M2 Macrophage	Promotes tumour growth, angiogenesis, suppresses T kill rate
NK Cell	MHC-independent tumour killing, boosted by pDC
Dendritic Cell	Phagocytoses tumour, activates T cells via antigen presentation
Plasmacytoid DC	Spawns NK, modulates angiogenesis, T proliferation
Treg	Suppresses T kill rate, proliferation, activation; reduces DC phagocytosis
Mast Cell	Dual role: tumour apoptosis & tumour growth; angiogenesis; spawns DC
Th1	Spawns M1 & DC; promotes CTL activity
Th2	Spawns M1 & T cells; promotes humoral branch
Tumour Cell	Divides, mutates DNA, competes for glucose, undergoes apoptosis

3D Grid Environment

SharedGrid Configuration

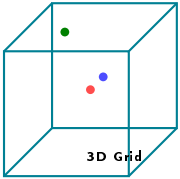
- `SharedGrid` — 3D, MPI-distributed
- `BorderType.Sticky` — clamped boundaries
- `OccupancyType.Multiple` — agents co-locate
- Grid dimension derived from tissue volume: $d = \lfloor (V/B^3)^{1/3} \rfloor$

Spatial Index

- `dict[(x,y,z)]` → `list[Agent]`
- O(1) lookup, $O((2r+1)^3)$ neighbour scan
- Moore (26-connected) or Von Neumann
- Updated on every `add/remove/move`

Deferred Movement

1. Agent sets `_desired_pos` in its `step()`
2. All agents finish stepping
3. `model.move_all_agents()` applies moves
4. Avoids conflicts during iteration



Simulation Loop — `model.step()`

1. **Collect data** — log counts, glucose stats
2. **Check termination** — max steps, tumour eradicated, tumour cap, growth threshold
3. **Apply effects** — neighbour-based effect collection
4. **Treatment** — ICI / TKI / ICI+TKI (after `treatment_start`)
5. **Sex drift** — CD8 population modulation
6. **Glucose step** — inject → diffuse → decay
7. **Angiogenesis** — add/refresh blood vessels
8. **Update search dimension**
9. **Shuffle & step all agents**
10. **Deferred move** — apply `_desired_pos`
11. **Spawn hormones** — from blood vessels

Termination Conditions

steps $\geq$ `max_steps` | tumour cells $= 0$ (survival) | tumour cells ≥ 2000 | mean growth rate $\geq$ threshold

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DNA Mutation Model

Genome Structure

- **16 genes** encoded as nucleotide strings
- Each gene: promoter (TATA box) + coding region + stop codon
- Full codon table (64 codons $\rightarrow$ 20 amino acids + stop)
- Wildtype protein = 20 amino acids per gene
- Total DNA sequence $\sim$3200 nucleotides

Mutation Mechanics

- **Point mutations** during `duplicate()`
- Mutation rate driven by **genomic instability** (BRCA1)
- Number of mutations: $n = \lfloor r_{\text{mut}} \times L \rfloor$
- Random nucleotide substitution (A, T, C, G)
- Protein similarity via **Levenshtein distance**
- Sex-dependent initial mutations: 3 (M) vs. 1 (F)

`MHC, B2M, CD274, BRCA1, TP53, RB1, CDKN2A, MYC, RAS, EGFR, HER2, JAK1, JAK2, TAP1, TAP2, APC`

Phenotypic Effects from DNA

Phenotype	Genes	Mechanism
Antigen Presentation	MHC, JAK1/2, B2M, TAP1/2	Loss of function $\Rightarrow$ immune evasion; computed as $\max(0,\ 6 - \bar{m}/6)$
Checkpoint Inhibition	CD274 (PD-L1)	Mutation $\times$ expression $\Rightarrow$ PD-1/PD-L1 suppression
Genomic Instability	BRCA1	Mutation $\text{score}^{\text{expr}} \times 0.05 \Rightarrow$ accelerating mutation rate
Tumour Suppression	TP53, RB1, CDKN2A	Expression $\times(1 - \text{mutation})$; averaged over 3 genes
Extra Proliferation	MYC, RAS, EGFR, HER2, APC	Oncogene expression drives additional divisions; sigmoidal mapping

Treatment System

ICI — Anti-PD-1/PD-L1

- Blocks PD-1/PD-L1 axis on tumour cells
- Restores T cell activation effect
- Increases immune infiltration (Th1, Th2, Mast)
- Effectiveness $\propto$ `w_ici_effectiveness`

Combination: ICI + TKI

- Each drug applied with proportion $= 1/N_{\text{drugs}}$
- Effects are **proportionally averaged**
- Synergy: TKI releases neoantigens $\rightarrow$ ICI boosts T cell response

TKI — Anti-VEGF / Tyrosine Kinase

- Reduces tumour growth effect: $g \leftarrow g \times (1 - e)$
- Reduces angiogenesis effect: $a \leftarrow a \times (1 - e)$
- Releases neoantigens (`TKI_effect`)
- Reduces Treg differentiation
- Effectiveness $\propto$ `w_tki_effectiveness`

Treatment starts at step `treatment_start` (default: 100)

Sex Hormone System

Three Hormones

- **Estrogen** (weight 0.45)
- **Testosterone** (weight 0.30)
- **Progesterone** (weight 0.25)

Sex-Specific Spawn Rates

Hormone	Male	Female
Estrogen	0.2	0.9
Progesterone	0.1	0.8
Testosterone	1.0	0.2

T Cell Modulation

- Each hormone level $\rightarrow$ low/mid/high tier
- Each tier maps to signed effect scores
- 17 effect channels (Th1/Th2 proliferation, IFN-γ, PD-1, CTLA-4, MHC, kill rate, ...)
- 10% per-step hormone decay

$CD8^+$ Sex Drift

- Female: population shrinks ($-$ drift)
- Male: population grows ($+$ drift)
- Max drift 25% over 15 steps

Effect System — 14 Fields

Effect Fields

1. `t_kill_rate_effect`
2. `t_proliferation_effect`
3. `t_apoptosis_effect`
4. `t_activation_effect`
5. `th1_proliferation_effect`
6. `th2_proliferation_effect`
7. `treg_differentiation_effect`
8. `macrophage_m1_mutation_effect`
9. `macrophage_m2_mutation_effect`
10. `angiogenesis_effect`
11. `tumour_growth_effect`
12. `tumour_apoptosis_effect`
13. `dc_phagocitosis_effect`
14. `nkl_kill_rate_effect`

Mechanism

- Each cell produces an `Effect` on neighbours
- Collection radius: `EFFECT_RADIUS` $= 3$
- Aggregation: **sum** of all neighbour effects
- BMI normalisation:

$$z_{\text{BMI}} = \frac{\text{BMI} - \mu_{\text{sex}}}{\sigma_{\text{sex}}}$$

- 4 BMI-modulated fields: Treg diff., M1/M2 mutation, NK kill rate
- Effects reset each step via fresh `Effect(model)`

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Mesa → Repast4Py Translation

Mesa	Repast4Py
`mesa.Agent`	`repast4py.core.Agent(id, type, rank)`
`self.model.random`	`model.rng` (`random.Random(seed)`)
`self.model.schedule`	Snapshot list + `rng.shuffle`
`grid.get_neighbors()`	`grid_utils.get_neighbors_3d(spatial_index, ...)`
`isinstance(a, Cls)`	`a.uid[1] == AgentType.X`
`model.schedule.add()`	`model.add_agent()`
`model.schedule.remove()`	`model.remove_agent()` + `_alive` flag
`MultiGrid` (2D)	`SharedGrid` (3D, sticky borders)
Class-level state	Moved to `RCCModel` instance
`DataCollector`	Manual `data_log` list + CSV writer

Code Architecture

Project Structure

```
rcc_repast4py/
  src/
    agents/   (18 agent classes)
      agent_types.py
      cell.py, t_cell.py
      tumor_cell.py, ...
    systems/
      dna.py, effect.py
      glucose_field.py
      grid_utils.py
    model/
      rcc_model.py
      observer.py, cell_adder.py
    parameters/
    treatments/
    learning/
    visualization/
  tests/
  config/
  run.py
```

Key Design Decisions

- `spatial_index`: `defaultdict(list)` for O(1) position lookups — Repast4Py does not expose neighbour queries
- `AgentType(IntEnum)`: integer IDs for fast type dispatch via `uid[1]`
- Class-level state (e.g. `tumor_blood_sources`) moved to `RCCModel` to avoid MPI serialisation issues
- Deferred movement pattern prevents iterator invalidation
- `Effect` system uses field summation (no message passing)

Glucose Field Implementation

Reaction–Diffusion Equation

$$\frac{\partial C}{\partial t} = D\,\nabla^2 C \;-\; \lambda\, C \;+\; S(\mathrm{x}) \;-\; Q(\mathrm{x})$$

- C glucose concentration
- D diffusion coefficient ($< 1/6$ for stability)
- λ natural decay rate
- $S(\mathrm{x})$ source term (blood vessels)
- $Q(\mathrm{x})$ consumption (tumour + immune)

Numerical Scheme

- NumPy 3D array: `np.float64`
- `scipy.ndimage.convolve` with 6-connected Laplacian kernel
- Neumann (zero-flux) BCs via `mode='nearest'`
- Stability: $D < 1/6 \approx 0.167$

Step Sequence

1. Blood vessels **inject** glucose
2. **Diffuse** via discrete Laplacian
3. Apply natural **decay**: $C \leftarrow C(1-\lambda)$
4. Agents **consume** in their own step

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Parameter Space — 114 Parameters

Model Parameters (4)

- `volume` — tissue volume (mL)
- `block_size` — voxel size (μm)
- `random_seed`
- `max_steps`

Patient Parameters (18)

- Sex (M/F), BMI
- Treatment type, treatment start
- 13 cell-type concentrations
- NK delta

Weight Parameters (92)

- 62 positive weights (log-uniform)
- 15 biases ($[-1, 1]$)
- 4 effectiveness/probability $[0, 1]$
- 1 growth threshold $[0.75, 0.99]$
- 1 integer receptor threshold $[0, 3]$
- 1 max drift
- **8 glucose parameters** (new):
  - diffusion, decay, source rate
  - tumour/immune consumption
  - growth/immune sensitivity
  - chemotaxis strength

Optuna Bayesian Optimisation

Setup

- **Sampler**: Tree-structured Parzen Estimator (TPE)
- **Trials**: 250
- **Repeats**: 1 per trial (stochastic seed)
- **Objective**: minimise mean squared error vs. clinical data

Objective Function

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^{N} \begin{cases} (\text{steps}_i - \text{OS}_i)^2 & \text{if outcome matches} \\ 500^2 & \text{otherwise (high penalty} \end{cases}$$

Clinical Data: ARON-1

- Real-world patient cohort
- Features: sex, BMI, treatment, concentrations
- Target: overall survival (OS) in steps
- Binary outcome: survival vs. death

Output

- Best parameter set saved to JSON
- Trial errors logged to CSV
- Convergence curve: best error vs. trial number

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Population Dynamics

Tumour Growth Curve

- Initial seeding: 10 tumour cells
- Exponential growth modulated by:
  - DNA-driven proliferation
  - Glucose availability
  - Immune pressure
  - Angiogenesis
- Growth plateau or eradication depending on treatment

Immune Response Dynamics

- Early NK & M1 response (innate)
- DC-mediated T cell priming
- CTL expansion (adaptive peak)
- Treg-mediated suppression
- $CD8^+$ sex drift effect
- Progressive exhaustion (`w_progressive_exhaustion`)

Tracked Metrics (per step)

17 population counts | 5 kill counters | 4 glucose statistics (mean, total, min, max)

Glucose Field Evolution

Temporal Phases

1. **Initial**: uniform concentration ($C_0 = 1.0$)
2. **Source establishment**: blood vessels inject glucose $\rightarrow$ local peaks
3. **Depletion zones**: tumour cells create glucose sinks (Warburg)
4. **Steady state**: balance of injection, diffusion, and consumption

Spatial Patterns

- Glucose-rich zones near blood vessels
- Glucose-depleted zones in tumour core
- Gradient formation: ∇C drives chemotaxis
- Immune cells follow glucose toward vessels
- Discretised gradient $\rightarrow (\pm 1, \pm 1, \pm 1)$ step

Key Parameters

D	0.01–0.16
λ	0.001–0.1
S	1.0–20.0
Q_{tumour}	0.5–10.0

Treatment Comparison

No Treatment

- Tumour grows unchecked
- Immune system eventually overwhelmed
- Glucose depleted in tumour region
- Worst outcome: early termination

TKI Monotherapy

- Reduces tumour growth & angiogenesis
- Releases neoantigens
- Reduces Treg differentiation
- Limited by immune exhaustion

ICI Monotherapy

- Restores T cell activation
- Partial tumour control
- Effective when tumour is immunogenic
- Limited by angiogenesis

ICI + TKI Combination

- **Best expected outcome**
- TKI slows growth + exposes antigens
- ICI restores immune killing
- Synergistic: addresses both evasion mechanisms
- Matches ARON-1 clinical superiority

Sex-Specific Outcomes

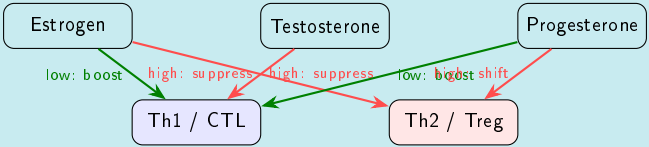
Male Patients

- Higher testosterone → suppresses Th1 at high levels
- Positive $CD8^+$ drift → more naïve T cells
- 3 initial BRCA1 mutations → higher genomic instability
- Higher incidence but poorer prognosis

Female Patients

- Higher estrogen → boosts Th1 at low levels
- Negative $CD8^+$ drift → fewer naïve T cells
- 1 initial BRCA1 mutation → lower instability
- Lower incidence, better immunotherapy response

Hormone-Mediated Mechanisms



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Statistical Validation: Multi-Seed Empirical Study

Experimental Design

- **160 simulations**: 20 random seeds $\times$ 8 scenarios
- 8 scenarios: 2 sexes (M/F) $\times$ 4 treatments (None, TKI, ICI, ICI+TKI)
- Each run: up to 300 steps, tuned parameters for visible treatment effects
- **Why?** Single-seed results can be misleading in stochastic ABMs

Statistical Methods

- **Survival rate**: fraction of runs where tumour cells $= 0$
- **Mann-Whitney U test**: non-parametric comparison of male vs. female outcomes
- Reported: mean $\pm$ standard deviation across 20 seeds

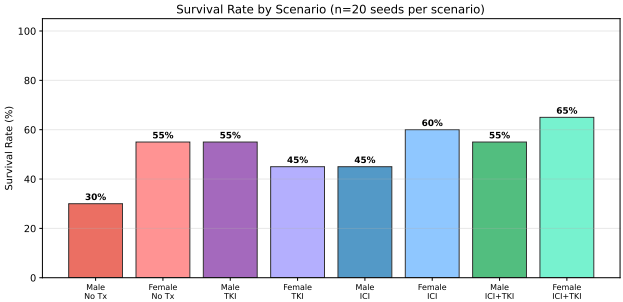
Empirical Results: Summary

Scenario	Survival	Tumour	Steps	Glucose
Male, No Tx	30%	578 ± 881	224 ± 88	3.10 ± 1.04
Female, No Tx	55%	257 ± 629	193 ± 91	2.73 ± 0.92
Male, TKI	55%	202 ± 596	209 ± 92	2.78 ± 1.23
Female, TKI	45%	205 ± 601	215 ± 92	2.74 ± 1.00
Male, ICI	45%	665 ± 920	199 ± 85	3.15 ± 1.28
Female, ICI	60%	274 ± 655	193 ± 86	2.81 ± 1.06
Male, ICI+TKI	55%	304 ± 716	192 ± 87	2.85 ± 1.28
Female, ICI+TKI	**65%**	$\mathbf{203 \pm 605}$	208 ± 81	2.81 ± 1.14

Key Finding

Female ICI+TKI achieves the **highest survival rate (65%)** — consistent with ARON-1 clinical observations of superior female response to combination immunotherapy.

Survival Rates by Scenario



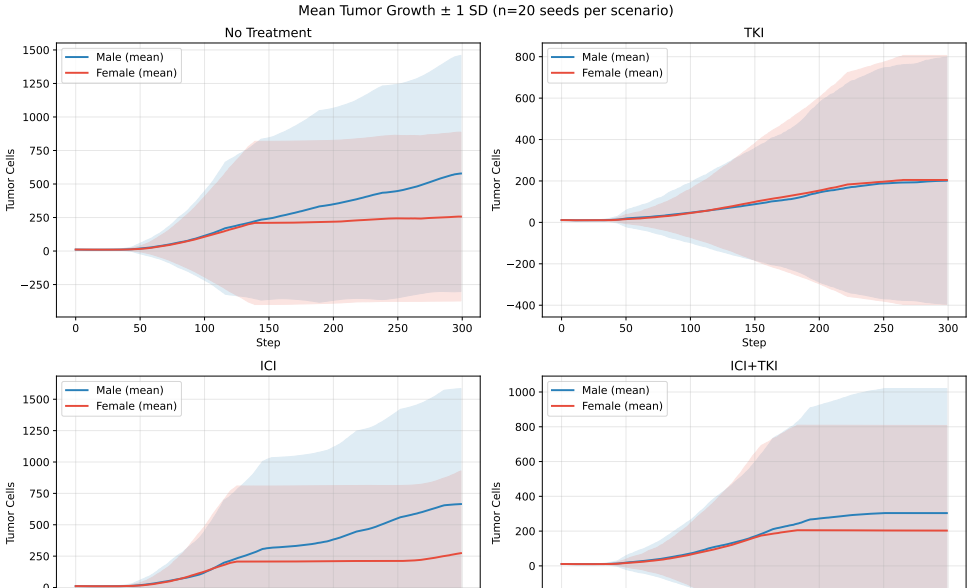
Treatment Effectiveness

- **No Tx**: M 30% vs F 55%
- **TKI**: M 55% vs F 45%
- **ICI**: M 45% vs F 60%
- **ICI+TKI**: M 55% vs F 65%

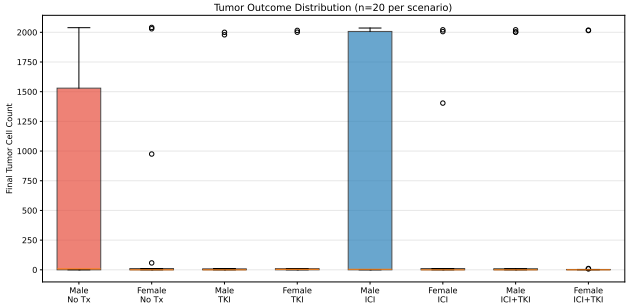
Observations

- Males benefit most from TKI (30%→55%)
- Females benefit most from ICI (55%→60%)
- Combination therapy best overall

Tumour Growth: Mean ± 1 SD (n=20)



Outcome Distribution



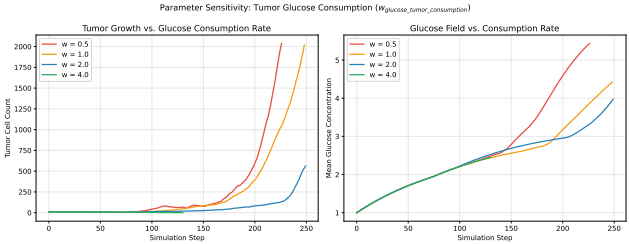
Statistical Tests

Mann-Whitney U (Male vs Female):

Treatment	p-value	Sig.
No Tx	0.139	ns
TKI	0.461	ns
ICI	0.230	ns
ICI+TKI	0.454	ns

Interpretation: Consistent trend favouring females, but $p > 0.05$ at $n = 20$. Larger sample sizes needed for significance.

Parameter Sensitivity: Glucose Consumption



Glucose Consumption Rate

w	Tumour	Outcome
0.5	2037	PROGR.
1.0	2020	PROGR.
2.0	561	PROGR.
4.0	**0**	**SURV.**

Interpretation

- Clear **tipping point** between 2.0 and 4.0
- Warburg effect creates meaningful metabolic constraint
- Validates glucose system as biologically relevant

Critical Insight: Single-Seed vs Multi-Seed

Single-Seed Result (seed=42)

- Female ICI+TKI: **PROGRESSION**
- 6 tumour cells remaining at step 299
- Appears to be the **worst** combination scenario

Multi-Seed Result (n=20)

- Female ICI+TKI: **65% SURVIVAL**
- Actually the **best** scenario overall
- Consistent with clinical evidence

Lesson

Single-seed analysis of stochastic models can be misleading.

Statistical validation across multiple seeds is essential for reliable conclusions in agent-based computational oncology.

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Summary

Achievements

- Successfully ported RCC ABM from **Mesa** → **Repast4Py** (MPI-based, 3D)
- Added **glucose microenvironment**: 3D reaction–diffusion with Warburg effect
- **18 agent types** with full immune system representation
- **16-gene DNA model** with codon translation, point mutations, 5 phenotypic effects
- **ICI + TKI treatment** system with proportional averaging
- **Sex hormone modulation**: 3 hormones, 17 effect channels, $CD8^+$ sex drift
- **92 tunable weight parameters** (including 8 glucose parameters)
- Bayesian optimisation with **Optuna** (TPE, 250 trials) vs. ARON-1 data
- **160-run empirical study**: 20 seeds × 8 scenarios with statistical validation
- Female ICI+TKI achieves **65% survival rate** — best scenario, consistent with clinical data

Technical Highlights

`spatial_index` dict for O(1) lookups | Deferred movement | `scipy.ndimage.convolve` diffusion | `AgentType` IntEnum dispatch

Future Work

Scalability

- **Multi-rank MPI**: distribute grid partitions across nodes
- **GPU acceleration**: offload glucose diffusion to CUDA/OpenCL
- Larger tissue volumes ($>10^6$ cells)

Biological Extensions

- Additional metabolites: **oxygen**, **lactate**, amino acids
- **Dynamic blood vessel remodelling** (angiogenesis/regression)
- Hypoxia-driven gene expression (HIF-1α)
- Cytokine diffusion fields (IL-2, IFN-γ, TGF-β)

Validation & Calibration

- **Extended validation** against larger clinical cohorts
- Multi-objective optimisation (survival + tumour burden)
- Sensitivity analysis (Sobol indices)
- Cross-validation over patient subgroups

Clinical Translation

- Patient-specific parameterisation from biopsy data
- Treatment sequence optimisation
- Digital twin for therapy planning